

VO₂ simulations: from working oscillator to experimental fit

Supporting automatic gain control in transimpedance amplifiers

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22 traces

Measured current waveforms
drive the simulations.

11 oscillators

Match which records oscillate,
and which remain stable.

One parameter set

Match voltage level, amplitude,
frequency and persistence.

Our goal is quantitative agreement across currents, not simply a waveform that oscillates.

First: establish the mechanism, then calibrate the sample

FOUNDATION / AUGUST 2026

Yuanhang-based control

Could the current-driven implementation sustain cycles without reaching zero volts?

What we changed

At $600\ \mu\text{A}$, increased C_{th} fourfold:
 $49.63 \rightarrow 198.51\ \text{pJ/K}$.

Other parameters retained Yuanhang's values.

Result: stable cycles at $0.222\ \text{MHz}$;
steady voltage range $0.906\text{--}6.333\ \text{V}$.

$V_{\text{floor}} \approx IR_m$: metallic resistance sets the steady floor;
 C changes the approach.

Replace control values with sample evidence

Input	Adopted value / basis
$R(T)$	Fit the measured major loop
R_m	$18.22\ \Omega$
T_0	$314.4\ \text{K}$
S_e	$3.675\ \mu\text{W/K}$, from $P/\Delta T$
C	$0.39\ \text{pF}$, timing upper bound
C_{th}	$0.04787\ \text{pJ/K}$, from $S_e\tau_{\text{th}}$
γ	0.95627 , initially borrowed

Thermal estimates depend on static $R(T)$ and circuit interpretation.
The major loop does not identify γ .

A working control verifies the mechanism; it is not a fit to our specimen.

Evidence: control 6765e0; resistance 0849a9; thermal estimates 761640 and aa2469; report Sections 1–6. Full run IDs and metric definitions in the accompanying README.

Gabriel Berezovsky / VO_2 simulation study

Trial 1: better waveform error did not recover oscillations

FROZEN PREDICTION → CONSTRAINED FIT → RELAXED FIT / 29 AUGUST

We replayed the measured current inputs. Then we fitted eight shared parameters using waveform, mean, amplitude, spectrum, frequency and edge errors.

Run	What changed	All-trace $J \downarrow$	Oscillating
Frozen estimates	No waveform tuning	0.8813	0 / 22
Constrained search	Stay inside supported intervals	0.8750	0 / 22
Relaxed diagnostic	Expand bounds; remove prior	0.6827	0 / 22

0.67 mV

Frozen-model mean-voltage error at the stable 189.6 μA record. Static agreement was good.

22.5% lower error

Relaxed fit improved J , but 7 of 8 parameters left their supported intervals. Only transients remained.

Reducing waveform error can reward a good average or a large transient, without producing sustained cycles.

Evidence: frozen eefab7; global fit 8f12d6; 17 training and 5 initially held-out records; final evaluation 0.05 ns. Full run IDs and metric definitions in the accompanying README.

Trials 2–4: trading oscillation detection against amplitude

CHANGE THE OBJECTIVE, THEN CONSTRAIN THE PHYSICS / 29 AUGUST – 7 SEPTEMBER

We successively emphasized oscillation detection, voltage amplitude, and independently supported parameters.

Run	Search strategy	Labels*	Amplitude MAE
Oscillation-priority	Reward detection; relax bounds	21 / 22	289.6 mV
Amplitude-priority	Increase amplitude penalty	19 / 22	81.2 mV
Physics-anchored	Anchor six supported parameters	22 / 22	212.5 mV

*Historical peak-count labels, not persistence. Amplitude MAE uses the same 11 experimental oscillators in every row.

What improved

We could generate cycles across many currents.
Amplitude weighting reduced voltage excursions;
anchoring limited parameter drift.

What remained unrealistic

The anchored fit needed $C = 6.84$ pF, about $17.5\times$ the timing bound, and $\gamma = 0.157$. Median amplitude was still $5.55\times$ measured.

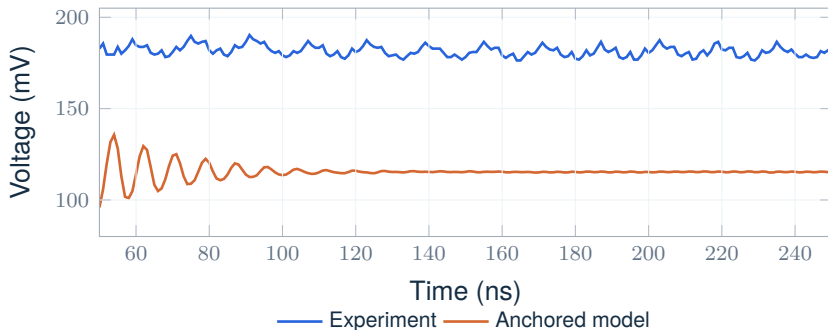
Each strategy improved a different target. None established a physically supported, quantitative fit.

Evidence: ac1c5e at 0.05 ns; 85526c and 717797 at 0.025 ns; objective values are not directly comparable across searches. Full run IDs and metric definitions in the accompanying README.

The turning point: some “oscillations” were dying out

PERSISTENCE AUDIT / 14 SEPTEMBER

606 μA : same measured current input



What we now measure

Four consecutive 50 ns windows: amplitude, mean, frequency and decay.

50.2 $\rightarrow$ 0.49 mV

Model raw peak-to-peak span, first to last window at 0.025 ns.

Experiment stays near 10–12 mV. The plotted fine-step replay shows the same failure.

At the finest step, the previous fit sustains only 9 of the 11 measured oscillators. The old 22/22 claim was too permissive.

Evidence: audit 26ff13; plot at 0.00625 ns; operational persistence thresholds, with sensitivity checks, are documented in report Section 11. Full run IDs and metric definitions in the accompanying README.

Trial 5: controlled parameter maps exposed the trade-off

126 SHARED COMBINATIONS / 228, 381 AND 606 μA

Vary C , $\tau_{\text{th}} = C_{\text{th}}/S_e$ and γ ; hold other parameters at the anchored fit.

Balance amplitude, frequency, mean voltage and decay. Recheck selected points on all 22 records.

Selected point	C (pF)	Recovered / 11	False +	Remaining failure
Best feature match	8.0	10	0	High-current cycles decay
Best persistent point	8.0	11	1	Amplitude far too large
Best within timing bound	0	0	0	No sustained cycles

294.5 vs 9.25 mV

Persistent-point vs measured late amplitude at 606 μA : $31.8\times$ too large.

No shared fit

No mapped point with $C \leq 0.39$ pF sustains cycles at any of the three selected currents.

Preserving cycles is not enough: the same parameters must also match their size and the non-oscillating records.

Evidence: 26ff13; table at 0.00625 ns; amplitude = 5–95% span over 150–250 ns; grid slices are not an exhaustive search. Full run IDs and metric definitions in the accompanying README.

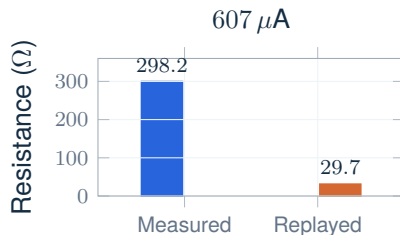
Trial 6: test the resistance law without another optimizer

CONDITIONAL RECONSTRUCTION FROM MEASURED POWER / 14 SEPTEMBER

Infer a thermal history from measured current and voltage, then replay the hysteresis law:

$$I_R = I - C\dot{V}, \quad R_{\text{eff}} = V/I_R, \quad C_{\text{th}}\dot{T} = VI_R - S_e(T - T_0).$$

Current (μA)	Inferred T (K)	Measured R_{eff} (Ω)	Replayed R (Ω)
189.6	330.91	1680.6	1679.2
228.7	328.96	1017.9	1762.5
381.1	333.64	488.4	1215.1
606.7	344.25	298.2	29.7



At 607 μA , four tested γ values give only **26–34 Ω** .
Even the upper S_e bound gives **71–112 Ω** , still below 298 Ω .

Changing γ alone did not close the gap. The inconsistency involves the resistance law, thermal assumptions or channel interpretation.

Evidence: fa4b66; means over 150–250 ns; central $C = 0.39$ pF and $\gamma = 0.95627$. Temperature is conditional, not independently measured; below-onset agreement is not independent validation. Full run IDs and metric definitions in the accompanying README.

What we learned – and the next decisive test

CONCLUSION / NO NEW CALIBRATED PARAMETER SET ADOPTED

Established by these trials

- 01 A working oscillator is not a sample fit.
- 02 Lower error and correct peak labels can conceal incorrect dynamics.
- 03 We have not found a shared, physically supported fit to amplitude, frequency and persistence.

Next: verify the measurement-to-model map

Identify where voltage was measured and how current was reconstructed.

Establish whether V/I is device resistance or includes contacts, load or readout response.

Then test one explicit correction at a time; validate on currents not used to tune it.

The next gain should come from testing an assumption, not merely increasing the optimizer budget.

Possible missing ingredients: dynamic or partial switching, nonuniform heating, or the actual TIA circuit. These remain hypotheses, not demonstrated causes.

Evidence: reviewed evidence through 14 September; all runs, source data and reproduction commands remain archived. Full run IDs and metric definitions in the accompanying README.